

Egypt

Digital agriculture diagnostic — the starting document for the National Digital Agriculture Roadmap (2026–2036).

Assessment: machine-assisted desk assessment Model: DAMM v1.7 Evidence compiled: 22 Aug 2026

Rendered: 22 Aug 2026 Indicators: 57 · 7 pillars × 4 layers



The evidence spine — one cell per indicator in census order: 20 Measured · 32 Documented · 0 Judged · 5 recorded gaps.

⚠ **Desk assessment, pending mission validation.** Machine-assisted desk assessment; 5 of 57 indicators are recorded gaps and 0 rest on judgment. No level exists without a recorded value; bands with weak evidence render parenthesized. Not for circulation, cross-country ranking, or financing decisions.

HOW TO READ THIS REPORT

Evidence classes

M Measured — a number was recorded **D** Documented — a citable artifact **J** Judged — assessor statement without artifact **G** Gap — searched for and not found. The class is derived from the recorded value, never chosen.

Pillars

A1 Agriculture & need • C1 Connectivity & access • C2 Data & digital public infrastructure (DPI) • C3 Policy & safeguards • C4 People & institutions • E1 Innovation, solutions & emerging tech • O1 Outcomes & inclusion.

Source tiers

T1 official statistics & international databases • T2 peer-reviewed & flagship reports • T3 government legal and policy artifacts • T4 reputable grey literature • T5 news/vendor material (admitted only in the register, for existence facts). Tiers are reported, never weighted.

Levels & bands

Every indicator carries a level, L1–L5. A pillar band is the plain mean of the levels actually recorded: Nascent <1.8 • Emerging <2.6 • Established <3.4 • Advanced <4.2 • Transformative ≥4.2. Recorded gaps and levels withheld pending ratification are not averaged in, so where a pillar has any, the count of rated rows is printed beside the mean. A band in (parentheses) rests more on judgment, gaps and withheld levels than on levelled evidence.

Use-case areas

ADV advisory & extension • SMF smart farming • MKT market linkage & pricing • SCM supply chain • FIN financial services • AGI agricultural intelligence (AI-enabled analytics and decision systems).

Playbook tags & IDs

“Feeds Playbook 1A...3C” ties each section to the step of the DAR Playbook (World Bank–Gates Foundation–BCG, 2025) it serves: 1A problem statement • 1B landscape • 1C ecosystem maturity • 2B use-case prioritization • 3C results baseline • 4A roadmap sequencing. Indicator IDs (e.g. 3.11) are the model’s census identifiers — all 57 resolve in the appendix; * marks a prerequisite.

C1 read *Advanced*; A1, C2, C3, C4, E1, O1 read *Established*. In C4 the band renders parenthesized — on current evidence, a finding about the assessment as much as the country. The assessment recorded values for 52 of 57 indicators and named 5 gaps; nothing here rests on an unrecorded claim. A further 5 carry a recorded value but no level: what was found measures a different construct from what the indicator names, so the level is withheld rather than guessed, and those rows are outside every mean.

STRONGEST PILLAR

C1 · 3.80

Connectivity & access — Advanced

WEAKEST PILLAR

O1 · 2.67

Outcomes & inclusion — Established

USE-CASE READINESS

0 R · 0 P · 1 B · 5 U

Ready · Partial · Blocked · Unverified — AGI blocked by 3.11, 4.5, 7.12 · 5 columns unresolved on prerequisite 2.1

EVIDENCE BASE

52/57

value-backed (M+D) · 47 levelled · 5 recorded gaps · 5 levels withheld · 11 stale

1 Country problem statement

FEEDS PLAYBOOK 1A

A1 is scored as need, not digital maturity — a low reading is a large opportunity. Pillar mean 3.00, Established, averaged over the 8 of 10 rows that carry a level (2 withheld pending ratification, 0 recorded as gaps); 10 of 10 are value-backed.

Agriculture value added per worker (USD)

8,942.4

M

Cereal yield (kg/ha)

7,402.2

ref: world ≈ 4,100 kg/ha · FAOSTAT 2023

M

Employment in agriculture (%)

17.5

M

L3 · World Bank WDI
SL.AGR.EMPL.ZS (ILO modeled)

Food production index (2014-16=100) **STALE**

109.1

M

L3 · World Bank WDI
AG.PRD.FOOD.XD · 2022

Prevalence of undernourishment (%)

9.4

ref: world 9.1% · FAO SOFI 2024

M

L3 · World Bank WDI
SN.ITK.DEFC.ZS · 2023

Women who own land (% holders) **STALE**

3.94

M

unrated · FAOSTAT, World Census of Agriculture domain — Egypt, item 'Civil holders, female', element 'Percent of total persons (holders)', Agricultural Census 2009/10 (CAPMAS) · 2010

Recorded from documents — 3 of 10 need metrics

- Post-harvest loss rate (%) (1.5 · L4)
- Smallholder access to formal markets (%) (1.6 · L1)
- Agricultural credit access (% farmers) (1.7 · L1)

D

levelled from cited evidence, not a national statistic

Recorded but unrated — 2 of 10

- Farmers using climate-smart practices (%) (1.8)
- Women who own land (% holders) (8.5)

D

evidence exists; it measures a different construct, so no level is set

Proposed additions to the need profile — recorded for context, not scored: **Cereal import dependency ratio**: 42.8% (2023, FAOSTAT, Suite of Food Security Indicators (FS domain), item 21035 'Cereal import dependency ratio (percent) (3-year average)', Egypt, period 2021-2023) · **Irrigation coverage**: 94.2% (2023, FAO AQUASTAT Dissemination System — Egypt, variable 4331 '% of the cultivated area equipped for irrigation', 2023 (flag E = estimate); companion variable 4313 'Area equipped for irrigation: total' = 3,823,000 ha (2023, flag I = imputed))

Reference values compare the country reading with the world figure for the same series — context only, never scored.

RESEARCHED NARRATIVE · DESK DRAFT — CITES SCORED ROWS; SETS NO LEVEL

Egypt's need profile inverts the usual reading: productivity is not the binding constraint. Cereal yields of 7,402kg/ha (1.2) run at close to twice the world average, on land where 94.2% of the cultivated area is equipped for irrigation (proposed indicator, not scored), and agriculture value added per worker stands at \$8,942 (1.1). The need

concentrates instead in exposure and inclusion: cereal import dependency of 42.8% (proposed indicator, not scored) leaves the food system exposed to world markets, and the smallholder interface is thin — 6% contract-farming participation (1.6) and 2.3% rural formal borrowing (1.7), both below the lowest band the model defines.

What Egypt measures, it measures well: 20 indicators carry national statistics and 32 more rest on citable artifacts, with no row resting on unevidenced judgment. Five are recorded gaps — the gender balance of the digital-agriculture workforce (5.12), digital adoption among agribusiness SMEs (6.13), climate-advisory reach (8.17), a business-environment score for which Egypt's assessment cohort has not yet published (6.3), and rural mobile broadband coverage (2.1). That last one governs this report. No rural-specific coverage figure is published for Egypt by the regulator, the ministry or the ITU; the national figure is 99.8%, but a national figure does not evidence a rural prerequisite. Five indicators more carry evidence but no level, because what was found measures a different construct from what the indicator names — those are listed in the ledger and flagged for ratification.

2 Initiative & solutions register

FEEDS PLAYBOOK 1B

One row per initiative, tier-badged. The register is the presence ladder applied to the landscape.

INITIATIVE · LEAD	USE CASES	STATUS	SCALE (AS REPORTED)	RESULTS EVIDENCE	OVERLAPS	SOURCE
Farmer Card (Kart el-Fallah) smart card program MALR with MCIT; Meeza card rails (state fintech e-Finance)	farmer registration & ID subsidized input entitlements digital payments	● Operating	3.5m cards printed, 2m distributed by Dec 2021 (official); from May 2026 the card is the sole approved payment method for subsidized fertilizer at cooperatives nationwide (news-reported)	No independent (T1-T3) outcome evaluation found; coverage figures are government-reported	farmer registration, payments, input subsidy	T3 MCIT-MALR joint press release, 6 Dec 2021 (accessed 2026-08-22); corroborated by Amwal Al Ghad, 10 May 2026 (T5), https://en.amwalalghad.com/egypt-kicks-off-summer-fertiliser-season-with-full-switch-to-digital-payments/ Cumulative card count as of 2026 not found. Al-Ahram Weekly Nov 2021 article and e-Finance Agrinet platform page both returned 403/rejected on access date, so the operating platform behind the card could not be independently profiled.

INITIATIVE · LEAD	USE CASES	STATUS	SCALE (AS REPORTED)	RESULTS EVIDENCE	OVERLAPS	SOURCE
Digitized subsidized-fertilizer distribution ('the fertilizer system') MALR (agricultural cooperatives; Meeza/Farmer Card payment rails)	input subsidy governance digital payments farm holding records	● Operating	Nationwide; 2026 summer season fully cashless with 5,500 electronic POS across 27 governorates and 6.6m fertilizer bags in the first tranche (news-reported, T5)	Officials claim fairer distribution and reduced subsidy leakage; no T1-T3 verification found	farmer registration, payments, input subsidy	T3 MALR official website, 'The fertilizer system' highlight (accessed 2026-08-22); scale figures from Amwal Al Ghad, 10 May 2026 (T5), https://en.amwalaighad.com/egypt-kicks-off-summer-fertiliser-season-with-full-switch-to-digital-payments/ Anti-leakage effects are official statements, not evaluated results. Desk report attributed fertilizer governance to 'AgriMisr' — mischaracterized; see issues.
Hudhud smart farmer assistant (AI pest & advisory app) MALR + MCIT	AI pest/disease diagnosis from photos Arabic-language advisory content	● Operating	Soft-launched Dec 2021; still an active flagship in MCIT-MALR program reviews as of Jul 2024; no user/download figures published at any tier	None published	advisory content	T3 MCIT-MALR joint press release, 6 Dec 2021 (accessed 2026-08-22); continuity confirmed by MCIT press release, 25 Jul 2024, https://mcit.gov.eg/en/Media_Center/Press_Room/Press_Releases/67819; existence also noted in ICARDA 2023 (T2) Two official press releases (2021, 2024) plus an ICARDA report confirm existence and continuity; usage and accuracy claims could not be verified.
El-Mufeed in Food and Agriculture digital extension app (desk report's 'AI Mufeed' and 'FAO El-Mufeed' — one initiative) FAO Egypt with MALR	digital extension content extension-agent support rural women/NGO outreach	● Operating	Promoted to extension agents in all governorates (148 participants at Dec 2020 national rollout meeting); no download figures published	Egyptian Knowledge Bank journal evaluations (2023, El-Gharbia and Assiut) reportedly rated app quality medium-to-high (~85% of officials) with internet/hardware obstacles — EKB deep links did not resolve on access date, so this results claim ships unconfirmed T2-claimed, link unresolved	advisory content	T2 FAO Egypt news, 22 Dec 2020 (accessed 2026-08-22); launch page: https://www.fao.org/egypt/our-office/the-fao-digital-app-of--el-mufeed-in-food-and-agriculture/ (31 Mar 2020) Deduplicated: desk report listed 'AI Mufeed and DAIRS apps' and 'FAO El-Mufeed digital extension model' as separate initiatives; they are the same app. Last independently dated activity is 2023 evaluation literature; 2026 operational status not re-confirmed.
DAIRS (climate early-warning / agro-advisory system) Attributed to MALR (per ICARDA); lead not independently confirmed	climate early warning	○ Announced	Unknown	None	advisory content, early warning	T2 ICARDA 'Current Situation of Digital Agriculture in Egypt' (2023), MELSpace record (accessed 2026-08-22) Weakest entry retained: sole trace is a passing mention in ICARDA 2023 describing DAIRS as a climate early-warning system in development. No official MALR/MCIT source, app-store listing, or news coverage found. Kept because the source is T2; flag for drop if finds nothing.

INITIATIVE · LEAD	USE CASES	STATUS	SCALE (AS REPORTED)	RESULTS EVIDENCE	OVERLAPS	SOURCE
Cropping-pattern satellite/AI program + agricultural holdings automation (Digital Egypt, MCIT-MALR) MCIT + MALR	remote-sensing crop mapping yield forecasting pest detection holdings registry automation e-portal services (~20 services in trial, 2021)	● Operating	Deployed for six major crops across five governorates as of Jul 2024 (official)	None published	remote sensing / crop monitoring, farmer registration	T3 MCIT press release, 25 Jul 2024 (accessed 2026-08-22) This row also covers the desk report's generic 'MCIT Digital Egypt agricultural services' candidate; the umbrella program's other components (Farmer Card, Hudhud) have their own rows.
Enhancement of Water Productivity in Agriculture project, Qena governorate FAO + MALR; funded by the Government of Japan	irrigation modernization water accounting ICT solar-powered irrigation good agricultural practices	● Operating	Qena governorate; farmer/feddan figures not published on project page	None published	remote sensing / crop monitoring, advisory content	T2 FAO Egypt office project page, updated Apr 2024 (accessed 2026-08-22) Desk report's 'water accounting ICT' component only broadly corroborated (FAO water-accounting work referenced on the page); project-specific ICT detail, duration and scale not published.
IFAD Sustainable Agriculture Investments and Livelihoods (SAIL) project Government of Egypt / IFAD	rural financial services agricultural marketing community development	● Operating	New-lands settlement areas in Upper, Middle and Lower Egypt; 2014-2026	Supervision reports through Jan 2026 exist on the project page but were not reviewed for outcomes in this pass	farmer registration (project beneficiary registry), credit/finance	T1 IFAD project page, ID 1100001745, status 'Ongoing' (accessed 2026-08-22) Desk-report claim of 'digital components' could not be verified from the project page; the Jan 2026 supervision report PDF is linked there for follow-up.
GIZ Agricultural Innovation Project II (Income Generation for Small-Scale Farmers) GIZ on behalf of BMZ; partner MALR	online advisory services online market platforms agri-fintech linkage climate-resilient practices	● Operating	Upper Egypt; Nov 2023 - Nov 2029; farmer numbers not specified	None published for phase II	advisory content, market linkage	T4 GIZ project page (accessed 2026-08-22); phase I page; https://www.giz.de/en/projects/income-generation-small-scale-farmers-agricultural-innovation-project Digital components stated in GIZ's own project description (T4, admissible for existence/design); no independent evaluation found.
Mozare3 (agri-fintech & contract farming platform) Private — co-founded by Hussein Abou Bakr, launched 2021	contract farming farmer financing (in-kind input finance) supply-chain digitization export market linkage	● Operating	Farmer/feddan figures not independently published	None at T1-T3	farmer registration (own onboarding), credit/finance, market linkage, advisory content	T5 EgyptInnovate (MCIT-affiliated portal) success story, 24 May 2026 (accessed 2026-08-22); existence corroborated by UNDP Digital Agribusiness Map, Jul 2025 (T4), https://www.undp.org/sites/e/files/zskgke326/files/2025-07/undp_agribusiness_digital_map.pdf UNDP (2025) maps Mozare3 under 'Digital Farmer Financing & Contracting' in the Delta region — independent T4 existence confirmation. Funding figure is portal/vendor-reported.

INITIATIVE · LEAD	USE CASES	STATUS	SCALE (AS REPORTED)	RESULTS EVIDENCE	OVERLAPS	SOURCE
FreshSource (B2B fresh-produce marketplace & logistics) — Private — founded by Farah Amara and Omar Amara	B2B produce marketplace cold chain & post-harvest logistics price/demand data services	● Operating	Serves hotels, restaurants, retailers, e-commerce and hospitals; farmer/volume figures not independently published	None at T1-T3	market linkage, supply chain	T4 UNDP Digital Agribusiness Map: Agri Tech & Climate-Smart Agriculture in Egypt, Jul 2025 (accessed 2026-08-22); current operations corroborated by StartupHub.today editorial, 15 Aug 2026 (T5). https://www.startuphub.today/en/editorial/fresh-source-egyptian-startup-uses-tech-to-streamline-food-supply-chains-2846 The Aug 2026 profile misstates founding year as 2026 (commonly reported 2019) — illustrative of T5 unreliability; widely reported US\$3m Series A (2022) not re-verified in this pass, so no funding figure is recorded.
Zr3i (satellite crop monitoring; now agricultural carbon MRV) — Private — Egyptian agritech	satellite crop monitoring precision-farming advisory (climate, irrigation, soil) carbon MRV (current focus) climate-risk insurance (planned only)	● Operating	Not published; MENA-focused platform-as-a-service	None at T1-T3	remote sensing / crop monitoring, advisory content	T4 GSMA Mobile for Development organization profile (accessed 2026-08-22); current positioning per vendor site (T5). https://www.zr3i.com/en Desk report described 'crop insurance' as a service; GSMA records insurance as planned, not live. Vendor site (2026) shows pivot to carbon-market MRV ('Zr3i 4.0').
AgriTech4Egypt Innovation Challenge CGIAR Accelerate for Impact Platform (Alliance of Bioversity International & CIAT) with Berytech	innovation challenge / venture acceleration digital agriculture & precision farming focus area	■ Closed	Global call (applications Mar-Apr 2024); program ran May-Dec 2024	Number and progress of supported ventures not verified	innovation support	T2 CGIAR news, call for applications, 2024 (accessed 2026-08-22); implementer page: https://agritechchallenge.org/projects/agritech4egypt; partner page: https://berytch.org/programs/agritech4egypt/ Implementer page states the challenge supported innovators from May to December 2024; no later cohort announcement found, hence status Closed.
Wheat silo automation & grain storage digitization Ministry of Supply and Internal Trade / Egyptian Holding Company for Silos & Storage, with IBM and ACME SAICO	IoT storage monitoring grain supply-chain analytics stock-level dashboards	● Operating	22 silos automated by end-2021 with stated plans toward 400 sites (vendor-reported); modernization program and FY2026/27 budget approved per Jun 2026 news	Leakage-reduction and visibility claims are vendor/ministerial statements; no T1-T3 evaluation found	supply chain, storage monitoring	T5 IBM MEA newsroom, 26 Sep 2021 (vendor, existence facts only, accessed 2026-08-22); ongoing status per Daily News Egypt, 14 Jun 2026 (T5). https://www.dailynewsegypt.com/2026/06/14/egypt-advances-silo-modernisation-efforts-to-safeguard-strategic-grain-reserves/ All scale figures vendor-reported per protocol. State Information Service page carrying the PM's Mar 2026 grain-system statement could not be fetched (TLS certificate error) on access date.

Register compiled to the Source-Tier Protocol (22 August 2026); every entry carries a deep link, and Tier 5 rows evidence existence only. Status uses the presence ladder (Announced / Adopted / Operating / Closed). T5 rows evidence existence only; scale figures marked “vendor-reported” are claims, not findings. Register rows may serve as Documented evidence for E1/C4 indicators — cited by row.

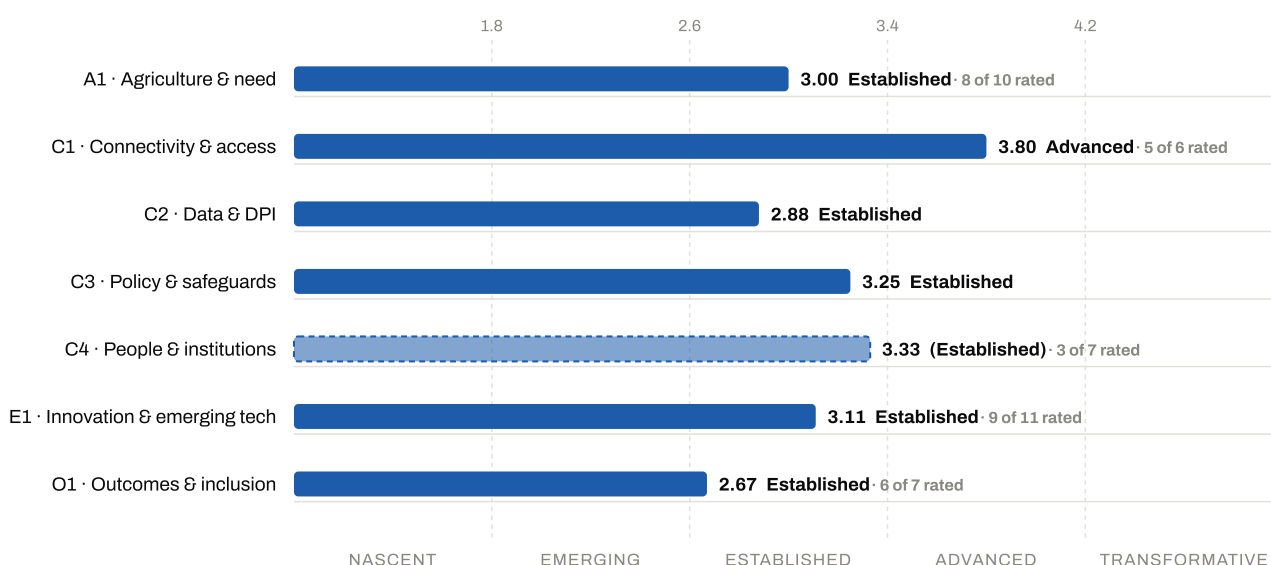
RESEARCHED NARRATIVE · DESK DRAFT — CITES SCORED ROWS; SETS NO LEVEL

Egypt's landscape is state-anchored: the Farmer Card is its spine — the mandatory channel for subsidized fertilizer, converted into a Meeza payment instrument — with FAO-linked digital extension (El-Mufeed), a state AI pest advisor (Hudhud) still at pilot reach, and a private cohort (Mozare3, FreshSource, Zr3i) that is real but thinly capitalized after the 2023 funding contraction.

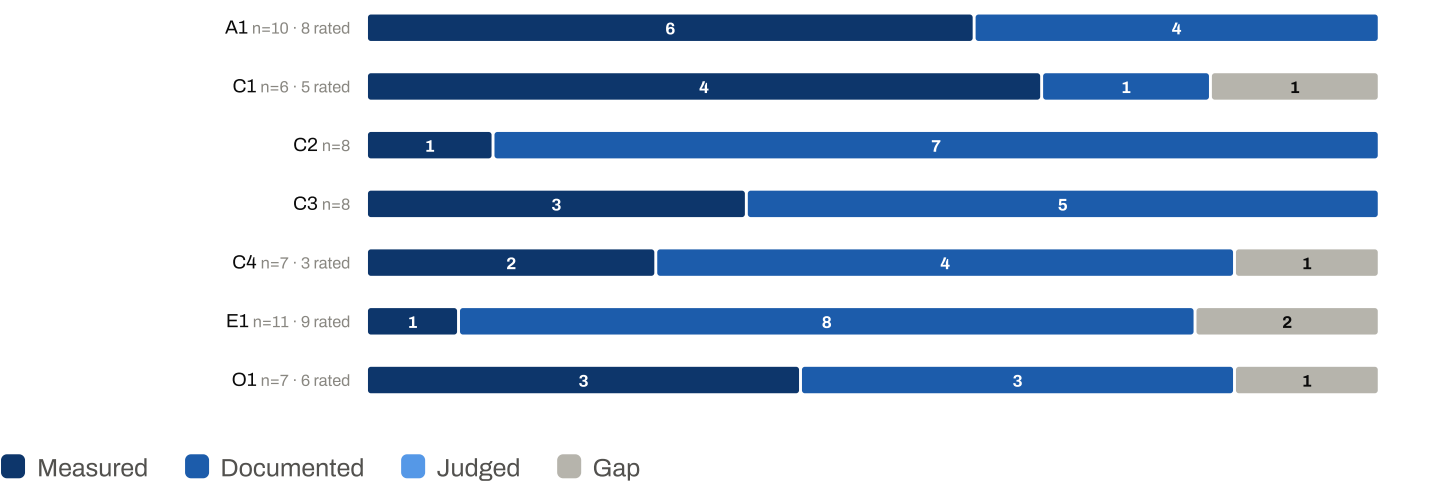
The dominant structural overlap is in ADVISORY/EXTENSION CONTENT: at least four public or donor channels deliver farmer advisory in parallel — Hudhud (MALR/MCIT, AI), El-Mufeed (FAO-MALR), DAIRS (announced), GIZ AIP II online advisory — plus private channels (Zr3i, Yalla Mozare3), with no evidence of a shared content backbone even between the two government-linked apps that target the same extension service. Second overlap: FARMER REGISTRATION/ID — the Farmer Card registry, the fertilizer system's holding records, the MCIT holdings-automation project, and each donor project (SAIL) and startup (Mozare3) maintain separate farmer/beneficiary registries; the Farmer Card is the only asset behaving as shared infrastructure (registry + Meeza payments) that another system (fertilizer distribution) demonstrably reuses. Third overlap: SATELLITE CROP MONITORING is being built at least three times — the MCIT-MALR cropping-pattern program, FAO water-productivity/water-accounting work, and Zr3i — with no documented data-sharing between them.

3 Pillar profile

FEEDS PLAYBOOK 1C



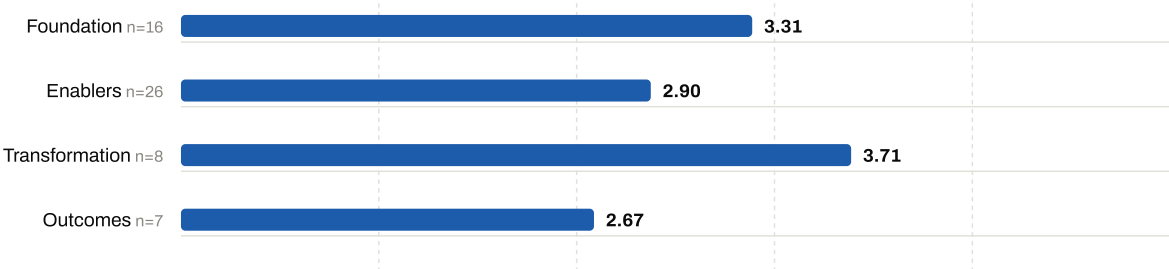
Band cut-offs at 1.8 / 2.6 / 3.4 / 4.2 (half-open). Hatched bars render parenthesized: judged and gap rows outnumber measured and documented ones, so the band is a finding about the assessment as much as about the country.



Evidence composition per pillar; the bar shows what class of evidence each row rests on, which is not the same as how many rows produced a level. A pillar mean averages only its rated rows, so A1, C1, C4, E1, O1 publish that count beside the mean. C4 carry the weak-evidence rendering: judged rows, recorded gaps and withheld levels outnumber the levelled measured and documented rows, so those bands print in parentheses.

4 Layer profile

FEEDS PLAYBOOK 1C



Leapfrog gap = mean(Foundation) – mean(Transformation) = -0.40 — no structural flag (flag at |gap| > 1.5). The national gap is calm; the use-case grain below is where fragility shows.

5 Use-case readiness matrix

FEEDS PLAYBOOK 2B — DIGITAL READINESS

Cells: Ready / Partial / Blocked (named blocker) / Unverified. Prerequisites bind on presence only — a fact, never an opinion. Delivery-risk flags sit on the cover and block nothing. The bearing set for a column is every indicator mapped to it, which includes agricultural-need and outcome indicators as well as enabling ones; both means are shown.

	ADV Advisory & extension	SMF Smart farming	MKT Market linkage & pricing	SCM Supply chain	FIN Financial services	AGI Agricultural intelligence
Readiness	? Unverified	? Unverified	? Unverified	? Unverified	? Unverified	× Blocked

	ADV Advisory & extension	SMF Smart farming	MKT Market linkage & pricing	SCM Supply chain	FIN Financial services	AGI Agricultural intelligence
Named blocker / reason	universal unverified: 2.1	universal unverified: 2.1	universal unverified: 2.1	universal unverified: 2.1	universal unverified: 2.1	3.11 Agricultural data interoperability standards, 4.5 Agricultural data governance framework, 7.12 Responsible-AI safeguards (consent, rights)
Mean of bearing indicators	3.27	3.12	3.13	3.22	3.03	3.17
Mean, enabling indicators only	3.30	3.21	3.35	3.33	3.25	3.26

UNIVERSAL — ABSENCE BLOCKS EVERY COLUMN

2.1

Rural mobile broadband coverage (3G/4G)

? Unverified

2.9

Rural electricity access (%)

✓ Present

4.1

Data protection/privacy law

✓ Present

PER USE CASE — ABSENCE BLOCKS NAMED COLUMNS

3.3

National farmer registry

FIN,AGI

✓ Present

3.11

Agricultural data interoperability standards

AGI

✗ Absent

4.5

Agricultural data governance framework

AGI

✗ Absent

4.7

Digital ID coverage (%)

FIN

✓ Present

5.5

Digital extension capability

ADV

✓ Present

6.14

Agri-fintech rails for smallholders

FIN

✓ Present

7.12

Responsible-AI safeguards (consent, rights)

AI

✗ Absent

DELIVERY-RISK FLAGS — ON THE COVER, BLOCKING NOTHING

4.9

Inter-ministerial coordination mechanism

✓ Present

5.7

MoAg digital/AI unit

? Unverified

ID	CONSTRAINT (A1 NEED PROFILE)	USE-CASE AREAS THAT ADDRESS IT
1.1	Agriculture value added per worker (USD): 8,942.42 (2025) M	SMF FIN
1.2	Cereal yield (kg/ha): 7,402.2 (2023) M	ADV SMF AGI
1.3	Employment in agriculture (%): 17.5246 (2025) M	SMF FIN
1.4	Food production index (2014-16=100): 109.07 (2022) M	ADV SMF
1.5	Post-harvest loss rate (%): No single national post-harvest loss rate exists. Cereals (labelled, admissible) D	SCM MKT
1.6	Smallholder access to formal markets (%): No national figure. Upper Egypt (regional, labelled): 6% of surveyed smallholder D	MKT
1.7	Agricultural credit access (% farmers): No direct 'of farmers with agricultural credit' figure exists for Egypt. Neare D	FIN
1.8	Farmers using climate-smart practices (%): No all-practice national CSA rate. Practice-specific (labelled): raised-bed whea D	ADV AGI
8.1	Prevalence of undernourishment (%): 9.4 (2023) M	ADV FIN MKT
8.5	Women who own land (% holders): 3.94 (2010) M	ADV SMF MKT SCM FIN AGI

Mapping: standard v1.7 crosswalk, curated by the task team leader (TTL). Constraints recorded as gaps still map — an unmeasured constraint is a task for the evidence ledger, not a reason to drop the use case.

7

Binding constraints

FEEDS PLAYBOOK 2B · 4A

Every indicator at L1–L2, ascending. * marks prerequisites — constraints that also gate a readiness column.

ID	INDICATOR (* = PREREQUISITE)	PILLAR	LEVEL
1.6	Smallholder access to formal markets (%)	A1	L1
1.7	Agricultural credit access (% farmers)	A1	L1
2.11	Device financing/subsidy schemes	C1	L1
3.11	Agricultural data interoperability standards *	C2	L1
3.8	National soil map/database	C2	L1
4.4	National digital agriculture strategy	C3	L1
4.5	Agricultural data governance framework *	C3	L1
6.1	Global Innovation Index	E1	L1
7.12	Responsible-AI safeguards (consent, rights) *	E1	L1
8.9	Smallholders reached by digital services (%)	O1	L1
3.5	Open agricultural data (machine-readable, AI-ready)	C2	L2
8.4	Mobile money account (%)	O1	L2

Q1 BLOCKED CELL AGI · 3.11

Agricultural data interoperability standards is Absent (3.11, L1). Which institution gets the mandate to move it to Adopted, on what timeline — and what does the agricultural intelligence agenda do until then?

PRECEDENT India's AgriStack published national agricultural data standards and registry schemas on digital ID rails — the standards mandate preceded the analytics ambition. T3

Q2 BLOCKED CELL AGI · 4.5

Agricultural data governance framework is Absent (4.5, L1). Which institution gets the mandate to move it to Adopted, on what timeline — and what does the agricultural intelligence agenda do until then?

PRECEDENT The EU Code of Conduct on agricultural data sharing (2018) shows a sector-specific governance instrument distinct from the general data-protection law. T4

Q3 BLOCKED CELL AGI · 7.12

Responsible-AI safeguards (consent, rights) is Absent (7.12, L1). Which institution gets the mandate to move it to Adopted, on what timeline — and what does the agricultural intelligence agenda do until then?

Q4 UNVERIFIED PREREQUISITE 2.1

The desk pass could not verify Rural mobile broadband coverage (3G/4G) (2.1) — a first-mission verification question, because a readiness column cannot resolve until it is settled.

Q5 UNVERIFIED PREREQUISITE 5.7

The desk pass could not verify MoAg digital/AI unit (5.7) — a first-mission verification question, because both delivery-risk flags hang on it.

Q6 UNIVERSAL PREREQUISITE 2.1 · AUTHORED

No rural mobile broadband coverage figure is published for Egypt at any admissible tier, so five use-case columns cannot be resolved either way. Is commissioning that single statistic — a rural split the regulator could publish from data it already holds — the cheapest unlock in this diagnostic?

Q7 STEP 2 SEQUENCING · AUTHORED

Setting the unverified prerequisite aside, the enabling profile is strong across the board and agricultural intelligence is the only column with named blockers. Which two use cases carry Vision 2030's food-security exposure — a 42.8% cereal import dependency — most directly?

Q8 EVIDENCE LEDGER · AUTHORED

Five indicators carry evidence but no level because the evidence measures a different construct from the indicator name, and five more are recorded gaps concentrated in outcome and inclusion metrics. Which does the mission commission first, and which does the model need to redefine?

Derived mechanically from the readiness matrix and the binding-constraints list; task team leader curation pending. Precedent pointers are existence proofs drawn from the DAMM Global Practice Library — relevance is selected by the readiness matrix; never endorsements, never country comparisons. Authored additions are marked as such.

9 Evidence ledger

FEEDS MISSION PLANNING

The assessment's honest edge: what to verify, what to refresh, before any of this is called a baseline.

Verify first 5

Judged rows and recorded gaps — the mission's evidence agenda.

C1 · Connectivity & access

2.1 Rural mobile broadband coverage (3G/4G) **G**

C4 · People & institutions

5.12 Gender balance in digital-ag workforce (%) **G**

E1 · Innovation, solutions & emerging tech

6.3 Business Ready (B-READY) **G**

6.13 SME/agribusiness adoption of digital tools (%) **G**

O1 · Outcomes & inclusion

8.17 Climate advisory reach (%) **G**

Refresh 11

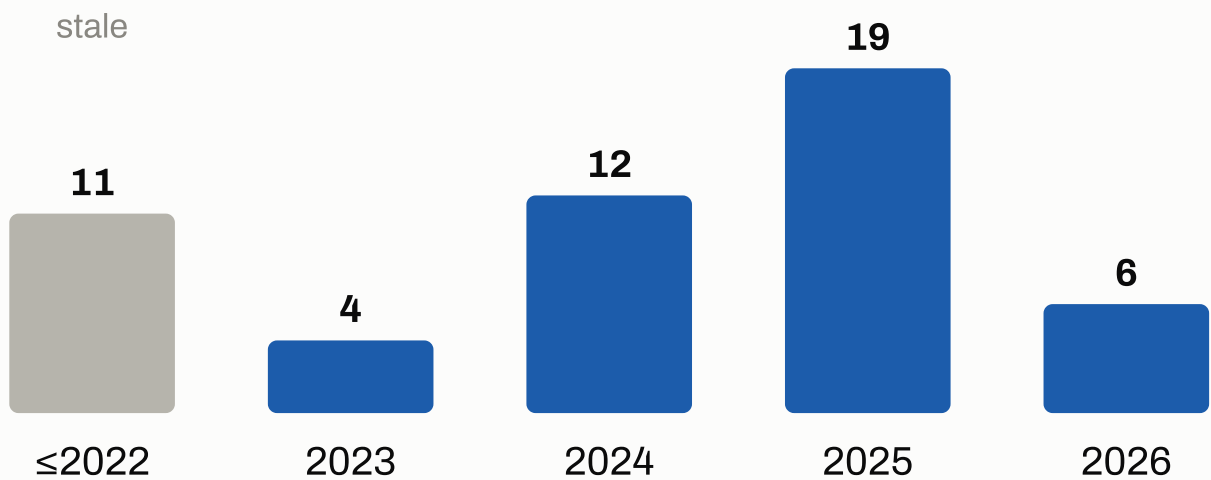
Measured, but older than the 3-year staleness rule.

1.4 Food production index (2014-16=100) (2022)

1.5 Post-harvest loss rate (%) (2019)

- 1.6 Smallholder access to formal markets (%) (2018)
- 1.7 Agricultural credit access (% farmers) (2018)
- 1.8 Farmers using climate-smart practices (%) (2017)
- 8.5 Women who own land (% holders) (2010)
- 2.7 Rural smartphone ownership (%) (2022)
- 4.4 National digital agriculture strategy (2021)
- 4.9 Inter-ministerial coordination mechanism (2021)
- 5.2 Adult literacy rate (%) (2022)
- 5.7 MoAg digital/AI unit (2021)

DATA VINTAGE



10 KPI baseline

FEEDS PLAYBOOK 30

ID	INDICATOR	BASLINE VALUE	YEAR	SOURCE	REFERENCE (CONTEXT)
1.1	Agriculture value added per worker (USD)	8,942.42	2025	World Bank WDI NV.AGR.EMPL.KD T1	—
1.2	Cereal yield (kg/ha)	7,402.2	2023	World Bank WDI AG.YLD.CREL.KG T1	world ≈ 4,100 kg/ha · FAOSTAT 2023
1.3	Employment in agriculture (%)	17.5246	2025	World Bank WDI SL.AGR.EMPL.ZS (ILO modeled estimate) T1	—
1.4	Food production index (2014-16=100)	109.07	2022	World Bank WDI AG.PRD.FOOD.XD T1	—
8.1	Prevalence of undernourishment (%)	9.4	2023	World Bank WDI SN.ITK.DEFC.ZS T1	world 9.1% · FAO SOFI 2024
8.5	Women who own land (% holders)	3.94	2010	FAOSTAT, World Census of Agriculture domain — Egypt, item 'Civil holders, female', element 'Percent of total persons (holders)', Agricultural Census 2009/10 (CAPMAS) T1	—

ID	INDICATOR	BASELINE VALUE	YEAR	SOURCE	REFERENCE (CONTEXT)
8.2	Account ownership, female (%)	40.2142	2024	World Bank WDI FX.OWN.TOTL.FE.ZS (Global Findex wave) T1	—
8.4	Mobile money account (%)	15.6	2024	World Bank, The Little Data Book on Financial Inclusion 2025 (Global Findex Database 2025), Egypt, Arab Rep. country page (p. 56): Mobile money account (% age 15+) = 15.6 T1	—
8.6	Gender gap in phone ownership (pp)	14.4	2024	World Bank Global Findex Database 2025, Egypt: Own a mobile phone, men (con1.2) = 91.7% vs women (con1.1) = 77.3%, age 15+, 2024 — gap 14.4 percentage points (derived by subtraction of the two published series) T1	—

Every Measured A1/O1 row, with value, source, year — the roadmap’s results framework starts from these six, and grows as the ledger converts gaps to measurements.

A All 57 indicators

FULL RECORD

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE · TIER (AUTO-DERIVED)
1.1	Agriculture value added per worker (USD)	A1	M	L4	8,942.42	2025	World Bank WDI NV.AGR.EMPL.KD T1
1.2	Cereal yield (kg/ha)	A1	M	L5	7,402.2	2023	World Bank WDI AG.YLD.CREL.KG T1
1.3	Employment in agriculture (%)	A1	M	L3	17.5246	2025	World Bank WDI SL.AGR.EMPL.ZS (ILO modeled estimate) T1
1.4	Food production index (2014-16=100)	A1	M	L3	109.07	2022	World Bank WDI AG.PRD.FOOD.XD T1
1.5	Post-harvest loss rate (%)	A1	D	L4	No single national post-harvest loss rate exists. Cereals (labelled, admissible): ~10% pre- and post-harvest losses (SADS 2030, estimate c.2009); wheat losses in traditional shona storage 10-20% (FAO 2019). Horticulture:	2019	FAO (2019), Food loss and waste in Egypt, CA4310EN, FAO Representation in Egypt T2
1.6	Smallholder access to formal markets (%)	A1	D	L1	No national figure. Upper Egypt (regional, labelled): 6% of surveyed smallholder households participate in contract farming (farm household survey Apr-May 2018, seven Upper Egypt governorates, n=2,237 households; IFPRI).	2018	Tabe-Ojong, M.P. Jr. & Abay, K.A. (2023), Smallholder farmers' participation in profitable value chains and contract farming: Evidence from irrigated agriculture in Egypt, IFPRI MENA Working Paper 42 T2

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE · TIER (AUTO-DERIVED)
1.7	Agricultural credit access (% farmers)	A1	D	L1	No direct '% of farmers with agricultural credit' figure exists for Egypt. Nearest labelled figures: only 2.3% of rural individuals aged 15+ had loans from formal financial institutions (ELMPS 2018, CAPMAS national panel)	2018	Mansour, S., Samak, N. & Gad, N. (2024). Credit Choices in Rural Egypt: A Comparative Study of Formal and Informal Borrowing. Journal of Risk and Financial Management 17(11): 487 T2
1.8	Farmers using climate-smart practices (%)	A1	D	—	No all-practice national CSA rate. Practice-specific (labelled): raised-bed wheat cultivation tripled 113,500 ha (2013/14) to 334,000 ha (2016/17) ~46% of wheat area; laser levelling and deficit-irrigation adoption documented	2017	USDA GAIN Egypt grain report (primary PDF); FAO/WB corroboration T4
8.1	Prevalence of undernourishment (%)	A1	M	L3	9.4	2023	World Bank WDI SN.ITK.DEFC.ZS T1
8.5	Women who own land (% holders)	A1	M	—	3.94	2010	FAOSTAT, World Census of Agriculture domain — Egypt, item 'Civil holders, female', element 'Percent of total persons (holders)', Agricultural Census 2009/10 (CAPMAS) T1
2.1	Rural mobile broadband coverage (3G/4G)	C1	G	—	DATA GAP — no rural-specific mobile broadband coverage is published for Egypt by ITU, NTRA or MCIT. Context (not scored): national population coverage at least 3G = 99.8% (ITU DataHub, 2025)	2025	Structured source search, 22 August 2026 T4
2.4	Individuals using the Internet (%)	C1	M	L4	74.6484	2024	World Bank WDI IT.NET.USER.ZS T1
2.5	Mobile broadband price (% GNI pc)	C1	M	L5	0.9	2025	ITU DataHub — Other ICT price baskets: Data-only mobile-broadband basket (2GB) (2021-2024 definition), % of GNI p.c. — Egypt (accessed 2026-08-22) T1
2.7	Rural smartphone ownership (%)	C1	M	L4	67.4	2022	MCIT ICT Indicators Quarterly Bulletin, June 2022 (Q2 2022), p.6 — Households with Smartphone according to (Urban/Rural)', reporting the MCIT/CAPMAS 'ICT usage in household and individuals' survey 2021-2022 T3
2.9	Rural electricity access (%)	C1	M	L5	100	2024	World Bank WDI EG.ELC.ACCS.RU.ZS T1

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE · TIER (AUTO-DERIVED)
2.11	Device financing/subsidy schemes	C1	D	L1	No government or telco device-financing/subsidy scheme targeting low-income or rural users found for Egypt (searched 2026-08-22). What exists instead: (1) commercial telco handset installments — Telecom Egypt 'WE Gold' p	2026	NTRA (tra.gov.eg) — 'Start of the installment service for fees due on mobile phones imported from abroad through consumer finance companies' (Arabic, 6 April 2026); NTRA — 'End of exceptional exemption on imported mobile phones' (21 Jan 2026) T3
3.1	UN E-Government Development Index	C2	M	L4	0.6699	2024	UN E-Government Knowledgebase — Egypt country profile (E-Government Survey 2024) T1
3.3	National farmer registry	C2	D	L4	National farmer registry EXISTS and is OPERATING: the MALR 'Farmer Card' (كارت الفلاح) smart-card system is the national electronic agricultural holding registry. MALR spokesperson (via SIS, 28 Mar 2026): ~5 million farm	2026	SIS (State Information Service): 'Agriculture spokesperson: continued efforts to ensure support reaches farmers through the Farmer Card' (Arabic), 28 March 2026 T3
3.4	Digital land/plot registration	C2	D	L3	Digital land registration is ADOPTED but not yet a comprehensive operating digital cadastre. New: Law No. 88 of 2025 (signed 18 June 2025; Official Gazette issue 24-bis, 17 June 2025) establishes a unified national elect	2025	SIS: 'President El-Sisi Signs Law Establishing National Real Estate Identification Number Database' (Law 88/2025), June 2025 T3
3.5	Open agricultural data (machine-readable, AI-ready)	C2	D	L2	Open agricultural data is ADOPTED in policy but the national open-data portal is NOT yet live. Egypt's first Open Data Policy (issued by the National Council for Artificial Intelligence, 2025; prepared by MCIT; reported	2025	Open Data Policy, Arab Republic of Egypt — issued by the National Council for Artificial Intelligence, 2025 (official PDF on ai.gov.eg) T3
3.6	Weather/climate data infrastructure	C2	D	L4	Weather/climate data infrastructure is OPERATING with ag-tailored services. The Egyptian Meteorological Authority (EMA) is the operational national met service (official site www.ema.gov.eg resolves; agrometeorology is a	2026	SIS: 'Agriculture issues its recommendations to farmers on dealing with the unstable weather wave in the coming days' (Arabic) (CLAC/ARC advisory) T3
3.7	Satellite/EO data integration	C2	D	L4	Satellite/EO integration in agriculture is OPERATING beyond pilots. T3 (MALR press conference, 6 Dec 2021): MALR and MCIT used AI to inventory the cropped area of each crop on the ground ('completed the summer-season cro	2024	MALR: 'Ministers of Agriculture and Communications announce pilot launch of digital transformation services in the agricultural sector' (Arabic), 6 December 2021 (includes completed AI crop-area inventory) T3

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE · TIER (AUTO-DERIVED)
3.8	National soil map/database	C2	D	L1	No operating national digital soil map/database service found — best-verified status is ABSENT for a public national digital soil information service, with legacy and research assets: FAO Soils Portal hosts legacy (digit	2026	FAO Soils Portal — Regional and National Soil Maps and Databases (Egypt legacy maps) T1
3.11	Agricultural data interoperability standards	C2	D	L1	No published agricultural data interoperability standards/schemas found for Egypt — ABSENT after active refutation attempt. Nearest instruments, none ag-specific: (1) Open Data Policy (NCAI, 2025, T3) mandates cross-gove	2026	Open Data Policy, Arab Republic of Egypt (NCAI, 2025) — closest instrument; no ag-specific standard exists T3
4.1	Data protection/privacy law	C3	D	L3	Personal Data Protection Law No. 151/2020 in force since October 2020; Executive Regulations issued by Decree No. 816/2025 of the Minister of Communications and Information Technology on 1 November 2025, published in the	2025	Shalakany Law Office, "New Executive Regulations for Egypt's Personal Data Protection Law" (Dec 2025) — full-text analysis of Decree 816/2025 T4
4.2	Cybersecurity framework (ITU GCI)	C3	M	L5	100	2024	ITU Global Cybersecurity Index 2024 (5th edition) — Egypt in Tier 1 'Role-modelling' (score band 95-100); exact score 100/100 per NTRA announcement T1
4.3	Government AI Readiness Index	C3	M	L3	55.63	2024	Oxford Insights Government AI Readiness Index 2024 — Full rankings annex: Egypt total 55.63 (Government 68.98, Technology Sector 42.13, Data & Infrastructure 55.77) T4
4.4	National digital agriculture strategy	C3	D	L1	No dedicated national digital agriculture strategy identified as of August 2026. Digital agriculture exists as components inside broader frameworks: (a) the updated Sustainable Agricultural Development Strategy (SADS 203	2021	MCIT press release, 'ICT Minister, Agriculture Minister follow up on joint digital transformation projects', 9 June 2021 T3
4.5	Agricultural data governance framework	C3	D	L1	No agricultural data governance framework (sector-specific rules on agricultural data sharing, ownership or use) identified — refutation attempt failed. The general PDPL 151/2020 and its Executive Regulations (Decree 816	2026	OECD, Policies for the Future of Farming and Food in Egypt (agricultural policy review — no agricultural data governance instrument appears) T2

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE · TIER (AUTO-DERIVED)
4.6	National AI strategy with agriculture component	C3	D	L4	Egypt National AI Strategy, 2nd edition (2025-2030), unveiled January 2025 (1st edition 2021), explicitly includes agriculture: named among strategic sectors for AI integration ('agriculture, manufacturing, defense, educ	2025	Egypt National Artificial Intelligence Strategy, Second Edition 2025-2030 (official English text, ai.gov.eg, 16 Jan 2025) T3
4.7	Digital ID coverage (%)	C3	M	L5	98.8	2024	World Bank Identification for Development (ID4D) Data — ID ownership, total (% of population ages 15+), Egypt, 2024 observation: 98.78 (API value 98.7781785562663; source 89, indicator ID.OWN.TOTL.ZS; 2025 dataset update based on the Global Findex 2024 survey round) T1
4.9	Inter-ministerial coordination mechanism	C3	D	L4	No gazetted standing inter-ministerial committee for digital agriculture identified; what exists and is documented is a continuous de facto coordination mechanism between MALR and MCIT: recurring joint ministerial review	2021	MCIT press release, 9 June 2021 — ICT and Agriculture Ministers follow up on joint digital transformation projects (portfolio listed; MALR digital transformation unit created) T3
5.2	Adult literacy rate (%)	C4	M	L3	79.4532	2022	World Bank WDI SE.ADT.LITR.ZS T1
5.3	Tertiary STEM enrollment (% gross)	C4	M	—	38.0398	2025	World Bank WDI SE.TER.ENRR T1
5.4	Digital literacy among farmers (%)	C4	D	—	No national farmer-specific digital-literacy rate is published for Egypt. Best available evidence (regional, single-source): IFPRI RCT baseline phone survey, Minya & Beni Suef governorates, Jul-Sep 2023, n=3,332 farmers:	2023	Abdelaziz, F. & Abay, K.A. (2025). Digital literacy training to promote diffusion of digital agricultural tools to smallholder farmers: Evidence from a randomized intervention in Egypt. IFPRI Discussion Paper 2359 (baseline survey Jul-Sep 2023) T2
5.5	Digital extension capability	C4	D	L4	Government digital-extension channels are operating: El-Mufeed (FAO/MALR digital advisory app, launched 2020, content produced with ARC, presented to extension agents from all governorates) and Hudhud AI farmer assistant	2025	IFPRI Policy Note Dec 2025 (Digital Services Portfolio/El-Mufeed operating); UN Egypt/FAO-MALR program page T2
5.7	MoAg digital/AI unit	C4	D	—	A joint MALR-MCIT digital transformation program has been operating since 6 Dec 2021 (Hudhud AI assistant; agricultural e-portal with 20 services in trial operation; AI/satellite crop mapping; quarantine mechanization; d	2021	MCIT-MALR joint press release, 'Digital Transformation Services in Agriculture Announced' (6 December 2021) T3

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE - TIER (AUTO-DERIVED)
5.8	Agtech/data-science training pipeline	C4	D	L3	No dedicated agtech/digital-agriculture university degree or certificate program was identified for Egypt. Citable pipeline elements: (1) Digital Egypt Builders Initiative (DEBI), MCIT, launched Sept 2020 in the President	2024	MCIT, Digital Egypt Builders Initiative (DEBI) program page T3
5.12	Gender balance in digital-ag workforce (%)	C4	G	—	DATA GAP — searched WDI/ILO female-agricultural-employment series and MCIT/CAPMAS ICT gender indicators (T1: population-level only, no digital-ag workforce breakdown); FAO Country Gender Assessment of the Agriculture and Rural Sector - Egypt (2022) and FAO Status of Women in Agrifood Systems (2023) (T2: no extension or digital-ag workforce gender share); moa.gov.eg / MALR publications (T3: nothing); GFRAS Worldwide Extension Study Egypt profile and JICA Egypt Country Gender Profile 2018 (T4: extension-staff proxies only, underlying data 2011). Searched 2026-08.	2026	Structured source search, 22 August 2026 T4
6.1	Global Innovation Index	E1	M	L1	24.7	2025	WIPO Global Innovation Index 2025 (full report), GII 2025 rankings table p.18: "86 Egypt 24.7" T2
6.3	Business Ready (B-READY)	E1	G	—	DATA GAP — Egypt is not covered in B-READY 2024 (51 economies) nor in B-READY 2025 (interim edition, 101 economies). Egypt is listed as economy #16 of 71 in the Third B-READY Report cohort (2026 full edition), which is not yet published as of 2026-08-22. Searched: World Bank B-READY publications page; official 'List of Economies Proposed for Inclusion in the First Three Editions of the B-READY Report' (thedocs.worldbank.org); B-READY 2025 report record (documents.worldbank.org); B-READY Methodology Handbook Third Edition (Jan 2026) confirming the 2026 rollout is in progress.	2026	Structured source search, 22 August 2026 T4
6.4	Agtech venture ecosystem	E1	D	L3	Operating agtech venture ecosystem: Egypt is one of Africa's 'big three' agrifoodtech markets (with Kenya and Nigeria); 22 agrifoodtech deals totalling USD 26M in 2024 (+2,775% YoY after a depressed 2023), vs USD 112M ac	2024	AgFunder, Developing Markets AgriFoodTech Investment Report 2025 (Africa: investment by country p.49: top-funded companies pp.53-54; investor list p.80) T4

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE · TIER (AUTO-DERIVED)
6.9	Public-private partnerships in digital ag	E1	D	L4	Multiple operating digital-agriculture partnerships with named partners: 'Agri Misr' e-platform (launched 13 Jul 2021: MALR + Ministry of Planning + Agricultural Bank of Egypt + eFinance Investment Group/eAswaaq); FAO x	2025	IFPRI Policy Note (Dec 2025): Digital Agricultural Technology in Egypt — Insights from App Developers (Abdallah, Abdelhadi & Karachiwalla) T2
6.12	Digital public goods adopted	E1	D	L4	One registered digital public good is operating in Egyptian agriculture: El-Mufeed, the Egypt deployment of the FAO Digital Services Portfolio (a DPG listed on the Digital Public Goods Alliance registry), run with MALR s	2025	FAO e-Agriculture: Proud Contributor Since 2022 — FAO in the 2025 State of the Digital Public Goods Ecosystem Report (DPGA registry includes the FAO Digital Services Portfolio) T1
6.13	SME/agribusiness adoption of digital tools (%)	E1	G	—	DATA GAP — no SME/agribusiness-specific digital-tool adoption percentage exists for Egypt. Searched in order: (1) CAPMAS/MCIT ICT-usage-in-enterprises survey (2019/20, economy-wide only: >70% of surveyed enterprises use computers/internet — no agribusiness breakout), as summarized in OECD 'SME and Entrepreneurship Policy in Egypt' (SME Digitalisation chapter); (2) World Bank Enterprise Surveys Egypt — no agricultural digital-tool adoption indicator; (3) UNDP Digital Agribusiness Map Egypt (2025) — publishes regional agri-tech adoption estimates (Delta 15%, Northern Coastal 10%, Upper Egypt 5%, Desert 3%) but these are farm-level regional estimates, not SME/agribusiness firm-level adoption; (4) IFPRI Dec 2025 policy note and A2K4D/AUC Mar 2025 report — qualitative only; (5) ITU DataHub — no enterprise agribusiness series.	2026	Structured source search, 22 August 2026 T4
6.14	Agri-fintech rails for smallholders	E1	D	L4	Operating agri-fintech rails with documented smallholder reach: national Farmer Card converted to a Meeza payment card on ABE/e-finance rails — 4.2 million smart farmer cards issued (Minister of Agriculture statement, 7	2025	IFPRI Policy Note (Dec 2025): Digital Agricultural Technology in Egypt — Insights from App Developers (harvest-linked finance platform, 5,000-6,000 active farmers/month) T2
3.9	Digital advisory platforms at scale	E1	D	L4	Multiple digital advisory platforms operating, several above 10k users (platform-reported, compiled by IFPRI): Mahsoly and AIMorshed Poultry 100K+ users each; several other advisory platforms 10K+; farm-management platfo	2025	IFPRI Policy Note (Dec 2025): Digital Agricultural Technology in Egypt — Insights from App Developers T2

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE · TIER (AUTO-DERIVED)
3.10	Agricultural e-commerce platforms	E1	D	L4	Multiple agricultural e-commerce/marketplace platforms operating: Mahsoly (crops and land marketplace with El-Obour wholesale price feed; >100K reported users), Mahaseel, Khodarcum, Khodar Supply, Cropsa (aggregation/mar	2025	IFPRI Policy Note (Dec 2025): Digital Agricultural Technology in Egypt — Insights from App Developers T2
7.2	AI-enabled agricultural solutions deployed	E1	D	L3	AI-enabled agricultural solutions deployed but with limited verified reach: Hudhud (MALR x MCIT Arabic AI assistant: chatbot + photo-based pest/disease identification), pilot-launched Dec 2021, live on Google Play (publi	2025	A2K4D/American University in Cairo (MENA AI Observatory), AI Applications for Food Security in Egypt: Practices, Opportunities, and Challenges (Nagla Rizk, March 2025) T4
7.12	Responsible-AI safeguards (consent, rights)	E1	D	L1	No agriculture/AI-specific consent or farmer-data-rights safeguards exist beyond the general regime. General regime now operational: PDPL 151/2020 became enforceable in practice with Executive Regulations issued by Minis	2025	Egyptian Charter for Responsible AI v1.0 (NCAl, 2023) — official PDF, ai.gov.eg (full-text searched: no agriculture/farmer provisions) T3
8.2	Account ownership, female (%)	O1	M	L3	40.2142	2024	World Bank WDI, FX.OWN.TOTL.FE.ZS (Global Findex wave) T1
8.4	Mobile money account (%)	O1	M	L2	15.6	2024	World Bank, The Little Data Book on Financial Inclusion 2025 (Global Findex Database 2025), Egypt, Arab Rep. country page (p. 56): Mobile money account (% age 15+) = 15.6 T1
8.6	Gender gap in phone ownership (pp)	O1	M	L2	14.4	2024	World Bank Global Findex Database 2025, Egypt: Own a mobile phone, men (con1.2) = 91.7% vs women (con1.1) = 77.3%, age 15+, 2024 — gap 14.4 percentage points (derived by subtraction of the two published series) T1
8.9	Smallholders reached by digital services (%)	O1	D	L1	No national share with provenance. Regional (labelled, single-source): IFPRI RCT baseline, Minya & Beni Suef - digital ag service use among surveyed smallholders far below 10%	2023	IFPRI DP 2359 / RCT baseline (CGSpace) T2
8.11	Services in local languages (%)	O1	D	L5	No quantified share published; qualitative evidence indicates local-language delivery of flagship services is effectively universal. All major identified digital-agriculture services in Egypt are delivered in Arabic: Hud	2025	UN Egypt / FAO-MALR: 'FAO & MALR invite extension agents to use El Mufeed app and disseminate among targeted groups' (official program page; El Mufeed is Arabic-language); corroborated by IFPRI Egypt CSA policy note (Dec 2025) naming Bashaer and Nabta T3

ID	INDICATOR	PILLAR	CLASS	LEVEL	RECORDED VALUE	YEAR	SOURCE · TIER (AUTO-DERIVED)
8.12	Documented impact evidence (yield/income)	O1	D	L3	Rung facts. PRESENCE: one completed, published, independent impact evaluation of a digital-agriculture intervention in Egypt — Abdelaziz, F. & Abay, K.A., 'Digital literacy training to promote diffusion of digital agricu	2025	Abdelaziz & Abay (2025/2026), IFPRI DP 2359 / Food Policy — RCT of digital literacy training for smallholder digital-tool adoption, Egypt T2
8.17	Climate advisory reach (%)	O1	G	—	DATA GAP — searched (2026-08-22, in order): web search for EMA / CCIC / DAIRS advisory-reach statistics (none published); IFPRI Egypt CSA policy note Dec 2025 (CCIC issues daily climate warnings; CCIC-EMA 'Bashaer' platform delivers tailored climate-smart recommendations to farmers — no reach figures; FAO RNE panelist states 'communication gaps mean that many smallholders never receive' warnings); targeted search for Bashaer user/reach numbers (none); ICARDA digital-ag review abstract (DAIRS 'under development', no reach); FAO/WFP Egypt climate publications (project-level reach only, no national %). No percentage of Egyptian farmers receiving actionable climate advisories exists with provenance.	2026	Structured source search, 22 August 2026 T4

Method — DAMM v1.7 in five lines

1. Every indicator gets a value, a source, and a year — or a recorded gap. No level without a recorded value.
2. The value sets the evidence class: number → Measured · citable artifact → Documented · neither → Judged · looked-for-and-not-found → Gap. Derived, never chosen.
3. Pillars report a band and their evidence composition. No weights, no overall score, no stage.
4. Prerequisites bind on presence only; a failed prerequisite blocks its use-case column, not a global stage.
5. Source tiers (T1–T5) are reported, never weighted. Researched narrative cites scored rows; **no narrative claim sets a level.**

Automated quality control: 8/8 renderer checks passed (consistency, provenance, reconciliation and presentation — a failed check blocks the render). **Human gates:** G1, assessor confirmation of every machine-filled row — executed 22 August 2026, 18 derivations adjusted with recorded reasons; G2, independent peer review of all prerequisite and judged rows — executed 22 August 2026, all 12 prerequisite rows re-verified and confirmed; G3, task team leader sign-off against the four prohibitions — pending.

Prohibitions: no cross-country ranking · no band as a project development objective (PDO), disbursement-linked indicator (DLI), or disbursement condition · no automatic financing decisions · no public claim before human review.
 Rendered by the DAMM v1.7 engine and report pipeline from the scored indicator set and the initiative register for

Egypt; the same pipeline renders any country’s assessment. Source tiers are assigned by the DAMM Source-Tier Protocol lookup.

Abbreviations in recorded values and sources

Source and value strings in the tables are quoted verbatim as recorded; alphanumeric series codes (e.g. NV.AGR.EMPL.KD) are World Bank WDI series identifiers. **A4AI** Alliance for Affordable Internet · **AfDB** African Development Bank · **B-READY** World Bank Business Ready assessment · **BoA** Bank of Agriculture · **CBA** community-based advisor · **CBN** Central Bank of Nigeria · **CSA** climate-smart agriculture · **DCAS** Digital Climate Advisory Services · **DPI** digital public infrastructure · **EO** earth observation · **FAO** Food and Agriculture Organization of the United Nations · **Findex** World Bank Global Findex database · **FMAFS** Federal Ministry of Agriculture and Food Security · **FMARD** Federal Ministry of Agriculture and Rural Development (predecessor ministry) · **G2P** government-to-person payments · **GAID** General Application and Implementation Directive (NDPC, 2025) · **GII** Global Innovation Index · **GNI pc** gross national income per capita · **GSMA** the mobile-industry association · **ICR** Implementation Completion and Results Report (World Bank) · **IDA** International Development Association (World Bank) · **ILO** International Labour Organization · **ITU** International Telecommunication Union · **ITU GCI** ITU Global Cybersecurity Index · **KPI** key performance indicator · **LIFE-ND** Livelihood Improvement Family Enterprises in the Niger Delta (IFAD) · **MoAg** Ministry of Agriculture · **NAERLS** National Agricultural Extension and Research Liaison Services · **NAIS** National Artificial Intelligence Strategy (2024) · **NASRDA** National Space Research and Development Agency · **NBS** National Bureau of Statistics · **NCC** Nigerian Communications Commission · **NDAS** National Digital Agriculture Strategy (2020–2030 draft) · **NDPA** Nigeria Data Protection Act (2023) · **NDPC** Nigeria Data Protection Commission · **NiMet** Nigerian Meteorological Agency · **NIMC** National Identity Management Commission · **NIN** National Identification Number · **NITDA** National Information Technology Development Agency · **OECD.AI** OECD AI Policy Observatory · **SCP** Seasonal Climate Prediction (NiMet) · **SME** small and medium-sized enterprise · **STEM** science, technology, engineering and mathematics · **TTL** task team leader · **USSD** Unstructured Supplementary Service Data (feature-phone menus) · **WDI** World Development Indicators (World Bank) · **WIPO** World Intellectual Property Organization. **ARC** Agricultural Research Center · **CAPMAS** Central Agency for Public Mobilization and Statistics · **CLAC** Central Laboratory for Agricultural Climate · **DEBI** Digital Egypt Builders Initiative · **EMA** Egyptian Meteorological Authority · **MALR** Ministry of Agriculture and Land Reclamation · **MCIT** Ministry of Communications and Information Technology · **Meeza** Egypt's national payment card scheme · **NARSS** National Authority for Remote Sensing and Space Sciences · **NCAI** National Council for Artificial Intelligence · **NTRA** National Telecom Regulatory Authority · **PDPL** Personal Data Protection Law 151/2020 · **PDPC** Personal Data Protection Center · **SADS** Sustainable Agricultural Development Strategy 2030 · **SIS** State Information Service.